Interpret inequalities

1

a $n > 17$ **b** $n < 10$ **c** $w < \frac{2}{3}$ **d** $y \geq 12$

e $6x \leq -42$ **f** $5 + 4x \geq 20$

2

a p is less than 9. **b** f is greater than 14.

c m is less than or equal to 23. **d** a is greater than or equal to 88.

Solve inequalities

3 -2

4 20

5

a Is a solution. **b** Is a solution. **c** Is a solution. **d** Is not a solution.

6

a 10 **b** Yes

7

a 12 **b** Yes

8

a 27 **b** Yes

9

a Is not a solution. **b** Is a solution. **c** Is not a solution. **d** Is not a solution.

10 Yes, because we need to divide by a negative number.

11 She should have reversed the inequality symbol in step 3.

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12

a $y > 6$	**b** $x \leq -6$	**c** $x > -7$	**d** $x \geq 4$
e $x < 16$	**f** $x > 5$	**g** $x \geq 8$	**h** $x > 6$
i $x \geq 13$	**j** $x > 5$	**k** $x \leq -7$	**l** $x \leq -3$
m $k \geq 4$	**n** $x > 2$	**o** $x \geq 21$	**p** $x \leq -32$
q $x \geq 63$	**r** $x < -6$	**s** $x \geq 6$	**t** $x > 0$
u $a < 66$	**v** $a > 0$	**w** $k \geq 9$	

13

a $p < 5$	**b** $x \geq 4$	**c** $x \geq 1$	**d** $x \leq 10$
e $x > -\frac{3}{4}$			

15

a $p \leq 4$ **b** $p = 4$

16

a $x \leq 1$

b

i Yes	**ii** No	**iii** Yes	**iv** Yes

17

a $x > 0$

b

i Yes	**ii** No	**iii** Yes	**iv** Yes

18

a $x \geq 3$

b

i No	**ii** Yes	**iii** Yes	**iv** No

Applications

19 $T < -15$

20 $x + 760 \geq 890$

21 Neville and Vincent

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22 Tobias and Bob

23 $x \geq 87$

24 $p \leq \$27.90$

25

a $h \geq 46.31$

b 47 hours

26

a $w \geq 9.65$

b 10 windows

27

a $x + 520 \geq 740$

b No

28

a $f > 3m$

b Example answer: 14 female fish and 4 male fish.

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